863. METHANOTHERMOBACTER/METHANOBACTERIUM TRANSDUCTION MEDIUM

Basal salts solution	100.000	ml
Trace elements solution	10.000	ml
Sodium carbonate solution	24.000	ml
Cysteine solution	5.000	ml
Na_2S solution	5.000	ml
Degassed distilled H_2O	856.000	ml

Anaerobically dispense liquid minimal medium in 10-ml portions into 100-ml serum vials under an atmosphere of N_2/CO_2 (80:20). Seal vials with butyl rubber stoppers secured with aluminium crimp seals. Sterilize by autoclaving for 15 minutes at 121°C. Prior to inoculation, pressurize bottles to 2 x 10^2 kPa with H_2/CO_2. For solidified plating medium, add the following components per liter of liquid minimal medium:

Agar	15.000	g
Titanium citrate solution	16.000	ml

Add titanium(III) citrate solution to warm (60°C) agar medium after autoclaving. For preparation of solidified agar medium.

Basal salts solution:

KH_2PO_4	68.000	g
NH_4Cl	21.200	g
Resazurin	10.000	mg
Distilled or deionized H_2O	1000.000	ml

Trace elements solution:

Nitrilotriacetic acid (Aldrich N8407)	9.550	g
$MgCl_2$ x 6 H_2O	4.060	g
$FeCl_2$ x 4 H_2O	0.980	g
$NaWO_4$ x 6 H_2O	0.264	g
$NiCl_2$ x 6 H_2O	0.118	g
$Na_2Se_2O_3$	0.100	g
Na_2MoO_4 x 2 H_2O	0.024	g
$CoCl_2$ x 6 H_2O	0.023	g
Distilled or deionized H_2O	1000.000	ml

Adjust pH of nitrilotriacetic acid in solution to 7.0 with KOH. Add minerals and adjust the final pH to 7.0.

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Na_2CO_3 solution:
Dissolve 106 g Na_2CO_3 in 1 liter of degassed H_2O. Autoclave in sealed serum bottles under CO_2.

Cysteine and Na_2S solutions:
To 50 ml of boiled and cooled distilled or deionized H_2O add 5 g each of L-cysteine hydrochloride and Na_2S x 9 H_2O. Adjust pH to 10 with 3N NaOH and store under N_2 in a serum bottle sealed with a butyl rubber stopper.

Titanium (III) citrate solution (Zehnder and Wuhrmann 1976):

Titanium(III) chloride, 15% solution (Merck 808307)	5.000	ml
Sodium citrate, 0.2M	50.000	ml

Neutralize pH with a saturated sodium carbonate solution and store in a serum bottle under N_2.

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